

API Documentation

Overview

The **Volunteer Hours Management API** is a FastAPI backend for managing:

- users
- volunteers
- events
- shifts
- work logs
- CSV imports
- analytics
- project statistics

This API is designed for event organisers who need to track volunteer work accurately and generate useful summaries from stored data.

Base URL

Local development base URL:

`http://127.0.0.1:8000`

Interactive API docs:

- `/docs`

Homepage:

- `/`
-

Authentication

Most routes are protected and require a valid JWT access token.

Auth flow

1. Register a user with `POST /auth/register`
2. Log in with `POST /auth/login`
3. Copy the returned `access_token`
4. In Swagger `/docs`, click **Authorize**
5. Paste:

`Bearer YOUR_ACCESS_TOKEN`

Roles

Supported roles:

- admin
- organiser

Admin-only functionality

These routes are intended for admin users:

- import endpoints
-

Route Groups

- /
 - /auth
 - /volunteers
 - /events
 - /shifts
 - /work-logs
 - /imports
 - /analytics
 - /stats
 - /health
-

Endpoints

Home

GET /

Returns a simple homepage for the project with links to important routes such as /docs and /health.

Health

GET /health

Simple health check endpoint.

Response

```
{  
  "status": "ok"  
}
```

Authentication

POST /auth/register

Register a new user.

Example request

```
{  
  "email": "admin@test.com",  
  "full_name": "Admin User",  
  "password": "Test12345!",  
  "role": "admin"  
}
```

Example response

```
{  
  "id": 1,  
  "email": "admin@test.com",  
  "full_name": "Admin User",  
  "role": "admin"  
}
```

POST /auth/login

Authenticate a user and return a JWT access token.

Example request

```
{  
  "email": "admin@test.com",  
  "password": "Test12345!"  
}
```

Example response

```
{  
  "access_token": "your.jwt.token",  
  "token_type": "bearer"  
}
```

GET /auth/me

Return the currently authenticated user.

Response

```
{
  "id": 1,
  "email": "admin@test.com",
  "full_name": "Admin User",
  "role": "admin"
}
```

Volunteers

POST /volunteers

Create a volunteer.

Example request

```
{
  "volunteer_no": "V001",
  "name": "Ahmed Ali",
  "email": "ahmed@test.com",
  "phone": "772100000"
}
```

Example response

```
{
  "id": 1,
  "volunteer_no": "V001",
  "name": "Ahmed Ali",
  "email": "ahmed@test.com",
  "phone": "772100000"
}
```

GET /volunteers

List all volunteers.

GET /volunteers/{volunteer_id}

Get one volunteer by ID.

PATCH /volunteers/{volunteer_id}

Update a volunteer.

Example request

```
{
  "phone": "770000001"
}
```

DELETE /volunteers/{volunteer_id}

Delete a volunteer.

GET /volunteers/{volunteer_id}/hours

Return total worked time for one volunteer.

Example response

```
{
  "target_id": 1,
  "worked_minutes": 150,
  "worked_hours": 2.5
}
```

Events

POST /events

Create an event.

Example request

```
{
  "title": "Saudi National Day",
  "description": "Main event",
  "location": "Leeds",
  "event_date": "2025-09-27"
}
```

Example response

```
{
  "id": 1,
  "title": "Saudi National Day",
  "description": "Main event",
  "location": "Leeds",
  "event_date": "2025-09-27"
}
```

GET /events

List all events.

GET /events/{event_id}

Get one event by ID.

PATCH /events/{event_id}

Update an event.

DELETE /events/{event_id}

Delete an event.

GET /events/{event_id}/hours

Return total volunteer time recorded for one event.

Example response

```
{
  "target_id": 1,
  "worked_minutes": 480,
  "worked_hours": 8.0
}
```

Shifts

POST /events/{event_id}/shifts

Create a shift for an event.

Example request

```
{
  "name": "Morning Shift",
  "description": "Setup and preparation",
  "start_time": "2025-09-27T09:00:00",
  "end_time": "2025-09-27T12:00:00"
}
```

GET /events/{event_id}/shifts

List all shifts for an event.

GET /shifts/{shift_id}

Get one shift by ID.

PATCH /shifts/{shift_id}

Update a shift.

DELETE /shifts/{shift_id}

Delete a shift.

Work Logs

POST /work-logs

Create a work log.

Example request

```
{
  "volunteer_id": 1,
  "shift_id": 1,
  "checked_in_at": "2025-09-27T09:15:00",
  "checked_out_at": "2025-09-27T11:45:00"
}
```

Example response

```
{
  "id": 1,
  "volunteer_id": 1,
  "shift_id": 1,
  "checked_in_at": "2025-09-27T09:15:00",
  "checked_out_at": "2025-09-27T11:45:00",
  "worked_minutes": 150
}
```

GET /work-logs/{work_log_id}

Get one work log by ID.

PATCH /work-logs/{work_log_id}

Update a work log.

DELETE /work-logs/{work_log_id}

Delete a work log.

Worked-hours calculation

Worked time is calculated automatically using shift boundaries:

- `effective_start = max(checked_in_at, shift.start_time)`
- `effective_end = min(checked_out_at, shift.end_time)`
- `worked_minutes = max(0, effective_end - effective_start)`

This prevents over-counting outside the planned shift window.

Validation behavior

Examples of rejected cases:

- `checked_out_at` before `checked_in_at`
 - duplicate work log for the same volunteer and shift
-

Imports

These routes are intended for admin users.

POST /imports/volunteers

Upload a CSV file of volunteers.

POST /imports/events

Upload a CSV file of events.

POST /imports/attendance

Upload a CSV file of attendance/work-log data.

Flexible import behavior

The import system supports flexible spreadsheet structures.

Volunteers import The volunteers importer can work with alternate column names such as:

- `volunteer_no`
- `volunteer_id`
- `volunteer_number`
- `full_name`
- `volunteer_name`
- `email`
- `mail`
- `phone`

- mobile

It can also import volunteers when email or phone is missing, as long as a usable name exists.

Events import The events importer supports alternate column names such as:

- event_title
- title
- event_name
- event_date
- date
- location
- venue
- description
- details

It also supports more than one date format, including:

- YYYY-MM-DD
- DD/MM/YYYY

Attendance import The attendance importer supports spreadsheet-style attendance data and converts it into stored work-log information.

Example CSV files

Examples used in this project:

- volunteers_import_template_en.csv
- events_import_template_en.csv
- attendance_import_template_en.csv

Example import response

```
{
  "created": 10,
  "updated": 2,
  "skipped": 1
}
```

Analytics

GET /analytics/leaderboard

Return volunteers ranked by total worked hours.

Example response

```
[
  {
    "volunteer_id": 1,
    "name": "Ahmed Ali",
    "total_minutes": 600,
    "total_hours": 10.0
  }
]
```

GET /analytics/awards

Return volunteers grouped by award tier.

Default award tiers

- tier_a: 20 or more hours
- tier_b: 15 to less than 20 hours
- tier_c: 1 to less than 15 hours

Example response

```
{
  "tier_a": [],
  "tier_b": [],
  "tier_c": [
    {
      "volunteer_id": 1,
      "name": "Ahmed Ali",
      "total_minutes": 600,
      "total_hours": 10.0
    }
  ]
}
```

GET /analytics/volunteers/{volunteer_id}/summary

Return a volunteer summary.

Example response

```
{
  "volunteer_id": 1,
  "name": "Ahmed Ali",
  "total_events_worked": 2,
  "total_shifts_worked": 3,
  "total_minutes": 600,
  "total_hours": 10.0,
}
```

```

    "recent_activity": [
      {
        "event_id": 1,
        "event_title": "Saudi National Day",
        "shift_id": 2,
        "shift_name": "Evening Shift",
        "checked_in_at": "2025-09-27T17:00:00",
        "checked_out_at": "2025-09-27T19:00:00",
        "worked_minutes": 120,
        "worked_hours": 2.0
      }
    ]
  }
}

```

Stats

GET /stats

Return overall totals for the project data.

Example response

```

{
  "total_volunteers": 116,
  "total_events": 54,
  "total_shifts": 54,
  "total_work_logs": 116,
  "total_worked_minutes": 12000,
  "total_worked_hours": 200.0
}

```

Error Codes

Common error responses:

- 200 OK — successful read/update
- 201 Created — successful create
- 204 No Content — successful delete
- 400 Bad Request — invalid request or business-rule failure
- 401 Unauthorized — missing or invalid token
- 403 Forbidden — authenticated but not allowed
- 404 Not Found — requested resource does not exist
- 422 Unprocessable Entity — request validation error

Example validation error response

```
{
  "detail": "Validation failed",
  "errors": [
    {
      "loc": ["body", "email"],
      "msg": "value is not a valid email address",
      "type": "value_error"
    }
  ]
}
```

Example resource error response

```
{
  "detail": "Volunteer not found"
}
```

Notes

- The base path / now provides a homepage for the project
- /docs is the main interactive interface for testing the API
- Auth registration currently uses **full_name**
- Volunteer payload currently uses **name**
- Volunteer imports support files without email or phone
- Analytics are most meaningful after importing data or creating work logs manually